

For Module 4, I completed a Google Colab lab about Exploratory Data Analysis, also called EDA, using the Titanic dataset. The dataset included information about 891 passengers, such as whether they survived, their passenger class, sex, age, family information, fare, cabin, and port of embarkation. At first, I was a little confused when I started the notebook because there were a lot of commands and steps that I did not fully understand yet. I knew the lab was supposed to help me explore the data and find patterns before trying to build a model, but it took me some time to understand how the code, numbers, and graphs all connected together.

Once I realized that I needed to look closely at the code and fix the parts that were not working, I started having a blast with the lab. Figuring out how to correct the code made the assignment feel more hands-on instead of just running something that was already finished. I did not understand every command completely, but I was able to start recognizing when a new line of code needed to begin and how the code was organized. That was an important step for me because it showed me that I can still work through a notebook and troubleshoot problems even if I am not yet an expert with every Python command.

One part that stood out to me was loading and looking through the Titanic data with pandas. The notebook used commands like `df.head()` to display the first five rows and `df.info()` to show the column names, data types, and missing values. I thought it was interesting that the dataset had 891 entries but that some columns were incomplete. For example, the Age column only had 714 non-null values, and the Cabin column had much fewer available values. This showed me why it is important to inspect a dataset before making conclusions or trying to use it in a machine-learning model. If important information is missing, it can affect the results or make the data harder to work with.

I also liked using descriptive statistics to get a general overview of the dataset. The notebook showed statistics like the mean, minimum, maximum, and standard deviation for numerical columns. For example, the average passenger age was about 29.7 years, the overall survival rate was about 38.4%, and the fare amounts had a very wide range. Seeing the numerical summary was useful because it gave direct information about the passengers. However, I am a visual learner, so the graphs were the part I enjoyed the most.

Numbers give straight information, but graphs and visualizations feel like representations of that information. A table of numbers can tell me what happened, but a graph helps me see the pattern faster. The survival count plot was a good example because it made it easy to see that more passengers died than survived. The notebook explained that this can be considered an imbalanced dataset, which could matter later if the data were used for machine learning. I also found it interesting to compare survival with other

variables, such as passenger class, sex, fare, and port of embarkation. Visualizing those relationships made the data feel more real and helped me understand why EDA is useful.

Overall, this lab made exploratory data analysis much more interesting to me than I expected. At the beginning, I felt unsure because I did not understand every line of code. By the end, I enjoyed figuring out how to fix the code and seeing the results change when it worked. This assignment helped me practice reading a Google Colab notebook, working with a dataset, recognizing missing data, using summary statistics, and interpreting graphs. It also showed me that data analysis is not only about memorizing commands or looking at numbers. It is about exploring the data, asking questions, noticing patterns, and using visuals to better understand the information.